AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM240
Introduction to Computer Graphics

Midterm Examination (MTA)
Fall Semester 2025/2026

19/11/2025.

Date: 19/Nov./2025

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed:** 90

Instructions:

- Total Marks: **60**
- This exam consists of **2** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part -I: Question 1: Choose the best correct answer of all the following: (15 Marks)

1 .....is known as local or modeling coordinate system.

A Object space B World space
C Device space D View space

2 .....function is used to concatenate transformation to the right of the current transformation matrix.

A strcat B preConcatenate
C concatenate D cat

3 The space that holds all the graphics objects is commonly called .....

A World space B View space
C Object space D None of the above

4 An alpha-value is a number between 0.0 and 1.0 with .....representing complete transparency .

A 1 B 10
C 0.5 D 0

5 The center of the rectangle defined by RoundRectangle2D.Double(a1,a2,a3,a4,a5,a6) is at .....

A (a3,a4) B (a1+0.5a5,a2+0.5a6)
C (a5,a6) D (a1+0.5a3,a2+0.5a4)

6 The 2D cubic Bezier curve can be fully defined using .....points.

A 3 B 1
C 2 D 4

7 The function intersect is a member of the class .....

A Shape B Graphics2D
C Graphics D Area

8 The function Rectangle2D.double belongs to .....functions.

A Renderer B Builder
C Modeler D None of the above

9 .....is used to find the common points between two areas in java.

A Intersect B exclusiveOr
C Subtract D Add

10 .....is responsible for generating a visual image of a scene.

A operating system B renderer
C modeler D None of the above

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Part-II: Answer all of the following questions (45 Marks).

Question 2: (15 marks)

Having the following parametric equation:

$X(t) = 4\sin((3/4)t)$;

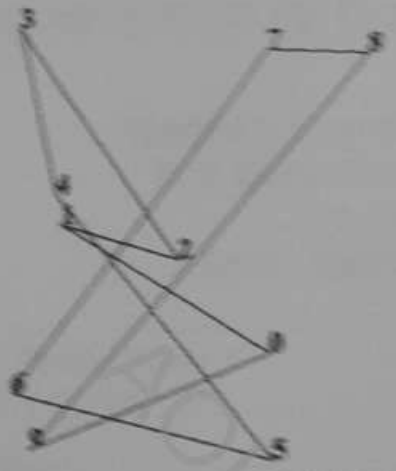
$Y(t) = 3\sin(t)$;

Write an appropriate Java code segment to plot the above function.

Question 3: (15 marks)

a. Describe the purpose and functionality of the TexturePaint class.

b. Having the following shape:



Copy the shape to your answer sheet then fill the interiors of this shape assuming Even odd winding winding rule and adopting the shown point order. (5 marks)

Question 4: (15 marks)

Having the following code along with its output:

```
g.drawString("Sample output from TM240 code.", 100, 100);
```

Sample output from TM240

Modify the above code to output this shape:

Sample output from TM240

End of questions